

Alperen Keles

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 alpaylan  Alperen Keles

Education

Doctorate of Philosophy - Computer Science 2021 - 2026
University of Maryland, College Park Maryland, USA
Thesis: **Designing Effective Property-Based Testing Frameworks**
Award: Certificate of Outstanding Achievement 2025-2026

Bachelor of Engineering - Computer Engineering 2017 - 2021
Middle East Technical University Ankara, Turkey
GPA: 3.66/4.0 (**top 5% in class of 229**)

Publications

Evaluating Shrinking (Experience Report) Haskell 2026
Alperen Keles, George Miao, Leonidas Lampropoulos

Programmable Property-Based Testing ICFP 2026
Alperen Keles, Justine Frank, Ceren Mert, Harrison Goldstein, Leonidas Lampropoulos

QuickerChick ICFP 2026 SRC
Ivan Mladenov, **Alperen Keles**, Leonidas Lampropoulos

DIRT: Database-Integrated Random Testing DBTest 2026
Alperen Keles, Ethan Chou, Harrison Goldstein, Leonidas Lampropoulos

Etna: An Evaluation Platform for Property-Based Testing JFP
Alperen Keles, Jessica Shi, Nikhil Kamath, Tin Nam Liu, Ceren Mert, Harrison Goldstein, Benjamin C. Pierce, Leonidas Lampropoulos

Tail: A Typed and Structured Document Editor POPL 2024 SRC
Alperen Keles

Etna: An Evaluation Platform for Property-Based Testing (Experience Report) ICFP 2023
Jessica Shi, **Alperen Keles**, Harrison Goldstein, Benjamin C. Pierce, Leonidas Lampropoulos

Protocol Verification Language CGO 2020 SRC
Alperen Keles, Ozan Akin, Ozan Sazak, Umut Sahin

DroPPPP: A P4 Approach to Mitigating DoS Attacks in SDN WISA 2019
Goksel Simsek, Hakan Bostan, Alper Kaan Sarica, Egemen Sarikaya, **Alperen Keles**, Pelin Angin, Hande Alemdar, Ertan Onur

Ongoing Research Projects

tjq (typed jq) Dec 2024 - Present
tjq is a gradual type system developed on top of jq, a json processing language.
(Gradual Types, Streams, Rust)

ETNA Sep 2022 - Present
ETNA is an evaluation and analysis framework for PBT libraries.
(Property-Based Testing, Rust, Python, Haskell, Rocq, OCaml, Racket)

Work Experience

Datadog Inc. New York, USA(Remote)
- Research Scientist II (expected start) Aug 2026

University of Maryland - PLUM Maryland, USA
- Graduate Research Assistant Aug 2022 - Jul 2026
Worked with Leonidas Lampropoulos on Property-Based Testing.

Datadog Inc. New York, USA(Remote)
- Research Science Intern Jul 2025 - Aug 2025
Worked on a performance fuzzing infrastructure for an LLM-guided continuous optimizer.
(Performance Fuzzing, AlphaEvolve)

Bloomberg L.P. - Research Intern Worked on developing executable semantics for inductive separation logic specifications in order to automatically derive property-based tests from separation logic propositions for testing C code. <i>(Separation Logic, Property-Based Testing, C, OCaml, Rust)</i>	New York, USA June 2024 - Aug 2024
Amazon Web Services - Applied Science Intern - Privacy Engineering Developed a dataflow analysis method for a state machine based DSL. <i>(Python, Java, Dataflow Analysis, Symbolic Execution, Abstract Interpretation)</i> - Applied Science Intern - Privacy Engineering Implemented of a novel encryption scheme on top of the AWS Encryption SDK. <i>(Python, Dafny)</i>	New York, USA May 2023 - Aug 2023 May 2022 - Aug 2022
University of Maryland - MC2 - Research Intern <i>(Program Synthesis, Automatic Exploit Generation)</i>	Maryland, USA(Remote) Jul 2020 - Dec 2020
Havelsan - Part-Time Engineer Designed and implemented a novel bandwidth detection and optimization algorithm for a video-conference application. <i>(Javascript, React)</i>	Ankara, Turkey Jul 2020 - Jul 2021
Emproof - Embedded Security Engineering Intern Worked on translation validation of binary obfuscation techniques. <i>(C++, Z3 SMT Solver, ARM Assembly, Symbolic Execution)</i>	Bochum, Germany Jun 2019 - Sept 2019
Community Service	
CAIS SAO Program Committee - Reviewer	May 2026
ICSE 2026 Shadow Program Committee - Shadow Reviewer	Aug 2025
extendr: Frictionless bindings for R and Rust - Reviewer	10.21105/joss.06394 Jul 2024
afetbilgi.com - Founder	Feb 2023 - Jul 2023
ICFP Programming Contest - Co-Organizer of 26th Annual ICFP Programming Contest - Organizer of 25th Annual ICFP Programming Contest	Sep 2022 - Jul 2023 Jun 2022 - Sep 2022
Talks	
The Art of Database Testing	DC Systems 011
Giving Types to jq	NJPLS May 2025
Property-Based Testing: The Past, The Present, and The Future	Bobkonf 25
Nitty Gritty Bits of Property-Based Testing	DC Systems 003
Selected Articles	
LLMs could be, but shouldn't be compilers	Feb 2026
Test, don't (just) verify	Dec 2025
A Better Vocabulary for Testing	Aug 2025
Verifiability is the Limit	Mar 2025